

MSTAR PRACTICE WORKSHEET**Lesson 2-7: Divide Decimals Using an Algorithm**

Grade 6 Mathematics · Standard 6.NOS.3

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: CWhat is $14.4 \div 1.2$?

- ☐ A 0.12
- ☐ B 1.2
- ☒ C 12
- ☐ D 120

Multiply both by 10: $144 \div 12 = 12$.**Question 2** SELECTED RESPONSE — CORRECT: CWhat is $3.78 \div 0.6$?

- ☐ A 0.063
- ☐ B 0.63
- ☒ C 6.3
- ☐ D 63

Multiply both by 10: $37.8 \div 6 = 6.3$.

Question 3 SELECTED RESPONSE — CORRECT: C

What is $8.4 \div 0.7$?

- ☐ A 0.12
- ☐ B 1.2
- ☒ C 12
- ☐ D 120

$8.4 \div 0.7 = 84 \div 7 = 12$.

Question 4 SELECTED RESPONSE — CORRECT: B

What is $15.6 \div 1.3$?

- ☐ A 1.2
- ☒ B 12
- ☐ C 13
- ☐ D 120

Multiply both by 10: $156 \div 13 = 12$.

Question 5 SELECTED RESPONSE — CORRECT: C

A water tank holds 19.2 liters. It is poured into bottles that each hold 1.6 liters. How many bottles can be filled?

- ☐ A 1.2 bottles
- ☐ B 11 bottles
- ☒ C 12 bottles
- ☐ D 13 bottles

$19.2 \div 1.6$: multiply both by 10 $\rightarrow 192 \div 16 = 12$ bottles.

Question 6

SELECTED RESPONSE — CORRECT: C

What is $4.8 \div 0.6$?

- ☐ A 0.8
- ☐ B 4.2
- ☒ C 8
- ☐ D 80

Multiply both by 10: $48 \div 6 = 8$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7**TWO-PART QUESTION****Part A — correct answer: C**

Find $18.9 \div 6.3$ using the decimal division algorithm.

- ☐ A 0.3
- ☐ B 2.8
- ☒ C 3
- ☐ D 30

Slide both decimal points one place right: $189 \div 63 = 3$. Multiplying both numbers by 10 leaves the quotient unchanged.

- **A:** The quotient was divided by 10 after the points were slid. Moving BOTH points the same direction leaves the answer unchanged.
- **B:** This division comes out even: $63 \times 3 = 189$, so the quotient is a whole number.
- **D:** The quotient was multiplied by 10 after the points were slid. 189 divided by 63 is exactly 3.

Part B — correct answer: D

Why does sliding BOTH decimal points the same number of places leave the quotient unchanged?

- ☐ A The decimal point does not affect division.
- ☐ B Only the divisor matters in a division problem.
- ☐ C The quotient always doubles when you slide a point.
- ☒ D Multiplying the dividend and the divisor by the same power of ten produces an equivalent division.

$18.9 \div 6.3$ and $189 \div 63$ are equivalent because both numbers were multiplied by 10 — the same scaling on both terms of a ratio keeps its value.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Moving Only One Point

Bianca computed $7.5 \div 0.5$. She slid the divisor's point to make it 5, but left the dividend as 7.5, and wrote ' $7.5 \div 5 = 1.5$ '.

Explain Bianca's misconception and give the correct quotient.

Model answer: $75 \div 5 = 15$. Bianca multiplied only the divisor by 10, which changes the problem. Both numbers must be scaled by the same factor, so 7.5 becomes 75. A check confirms it: $0.5 \times 15 = 7.5$.

2 points	Student explains that both decimal points must slide the same number of places to keep the division equivalent, computes $75 \div 5 = 15$, and notes that 1.5 is ten times too small.
1 point	Student gives 15 without explaining why both points must move.
0 points	Student agrees with Bianca or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A gas station sells fuel at \$3.45 per gallon. A customer pays \$27.60 total. How many gallons did the customer buy? Set up the decimal division, show how to convert it to whole-number division, and solve.

Model answer: Total cost divided by price per gallon gives the number of gallons, so the setup is $27.60 \div 3.45$. The divisor has two decimal places, so multiplying both numbers by 100 gives $2,760 \div 345 = 8$ gallons, and $8 \times \$3.45 = \27.60 confirms it.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.